Capstone Project - Financial News Sentiment Analysis

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Abstract

This report investigates the feasibility of stock price prediction using NLP. Presented is an overview of the issue, a comparison oif various textual sources, a literature review of the subject, and the detailing of development of a machine learning model to predict stock price movements, integrating sentiments and features extracted from smallcaps articles.

Introduction

need a proper introduction to the topic and THEN relating it to a data science problem

Predicting the movements of the stock market is a long standing challenge. Its erratic and volatile nature leads many to believe that predicting it is a fools game, with even experts in the field unable to consistently outperform the market [6–8]. The Efficient Market Hypothesis (EMH) claims that asset prices quickly reflect all available information [6], it follows that incorerating information and sentiment from sources across the internet may improve the performance of stock price prediction models to addess this age old problem. herd behavior and emotions can drive price swings beyond fundamentals [9].

Many efforts have been made to predict stock price movements, with a variety of methods employed, with machine learning being particularly popular in recent years [1].

Sentiment, particularly in recent times on the world wide web present a unique opportunity to leverage in this space. Online financial discourse is heavily concentrated across a few key domains, including social platforms (e.g., Reddit's r/WallStreetBets), news aggregators (e.g., Yahoo Finance), and article-based financial outlets (e.g., Bloomberg), and market announcements [10] . These domains offer continuous textual data that offer differing levels of up to date information and consumer sentiment.

A literature review conducted by [1] analysed 81 papers from January 2015 to June 2020, finding that support vector machines (SVM) were the most popular method, with 26.13% of the analysed papers using them, followed by random forest (RF) (15.32%), and long short term memory (LSTM) (13.51%)

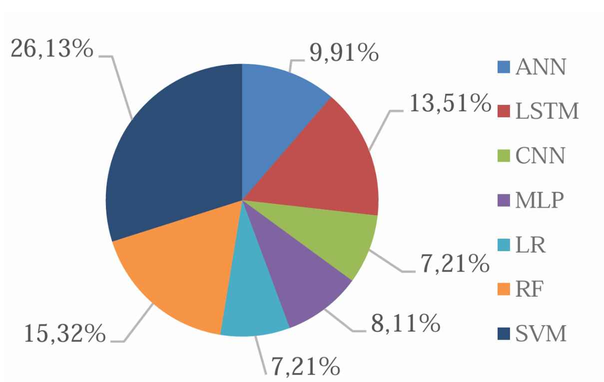


Fig. 1. Proportion of Each Stock Price Prediction method [1]

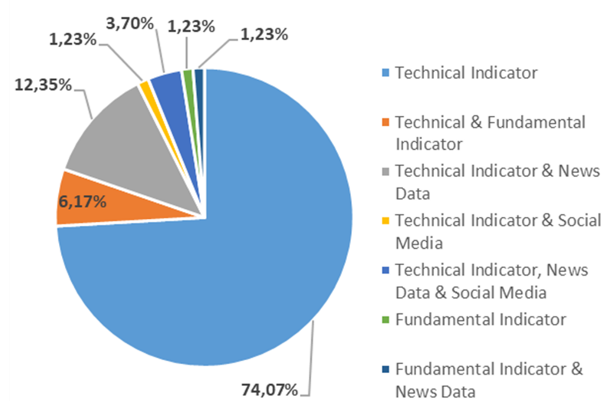


Fig. 2. Proportion of Each of the Studies' Data Source [1]

NLP techniques, not only have been proven applicable to the prediciton of stock price, but can also offer value through reducing the burden of manual newsd mmontoiring and analysis that is required by a day trader.

This report presents a pipeline that collects news articles using web scraping techniques and processes them through NLP systems to extract meaningful insights. All code and reporting files are available on GitHub.

Web Crawler

Data Type Selection

Several data sources were considered for this investigation.

The ASX market announcements [10] were initially examined. These are direct from the source and released frequently, but are often mundane or irrelevant. They are also provided in PDF format, which is not conducive to text mining using tools such as `BeautifulSoup`. Furthermore, usage is restricted under the ASX's terms of service [11].

Social media platforms, particularly Twitter and Reddit, were also considered. These offer a large volume of data, albeit with short, unstructured content. Posts commonly include colloquial language, spelling errors, requiring extensive preprocessing [12]; furthermore being less factual and trustworthy than other sources [13]. Unlike financial journalists, social media users are not held to standards of factual accuracy, limiting the reliability of this source.

In contrast, financial news articles are typically factual, well-written, and longer in length. Web-based sources often provide a more structured format and accessible metadata, making them better suited to text mining. It is for these reasons that financial news articles were selected as the primary data source for this investigation.

Website Selection

Table 1 shows the comparison of various websites that contain financial and market-related articles, evaluating their suitability for scraping. Copyright is a key legal consideration when using original written works. Under Section 113P of the Copyright Act 1968, Australian copyright law permits educational institutions to copy and communicate works for educational purposes under the 'fair dealing' provisions [14].

Websites were assessed based on their suitability for article scraping, with key criteria including explicit permission in terms of use, presence of disallow/allow directives in `robots.txt`, use of scraping inhibiting in the form of Cloudflare [15], and article structure (e.g. presence of stock codes or site metadata). As shown in Table 1, many domains impose legal or technical restrictions on automated data collection, presenting ethical limitations to their use. For example, scraping from websites that explicitly disallow it in their terms of service may breach legal and institutional ethical standards. From a sampling design perspective, sites with structured article metadata and consistent tagging (e.g. Trading Economics, Stock Head) were preferred to ensure replicable and unbiased data collection. Conversely, general news outlets with mixed content and unstructured articles (e.g. ABC News, Yahoo Finance) were excluded due to increased noise and ambiguity in topic association.

Small Caps was selected due to its static structure, enabling lightweight scraping using `BeautifulSoup`. All content is stock-focused, removing the need for filtering, and articles are consistently tagged with stock codes, simplifying parsing. However, using a single source limits dataset diversity and may bias coverage towards small-cap companies, or a particular sector, or reflect the opinions of a few authors.

Web Scraping

For the entirety of the scraping process, although the `robots.txt` file did not specify a scraping frequency, request throttling was implemented as a matter of ethical practice to minimise server load.

One merit contributing to the selection of *Small Caps* was its XML sitemap, which provides a consolidated list of article URLs. This removes the need to emulate page navigation with tools like `Selenium`. Instead, the sitemap is scraped using the `lxml` parser for its speed and ability to handle both XML and HTML. Table 2 shows the distribution of last modified dates for articles listed in the sitemap.

Website Name (URL hyperlinked)	**Scraping Permitted in Terms of Use?**	**Allows Article Scraping in Robots.txt?**	**Uses Cloudflare?**	**General Notes**
Market Index	No	Yes	No	Company tags on articles
Yahoo Finance	No	No	No	no mentioned stock labelling. General politics also included which is difficult to relate to individual stocks
Trading Economics	Yes	Yes	No	General politics and economics also included — would require filtering (unstructured)
Financial Review	No	No	No	
The Motley Fool	No	Yes	No	
Sydney Morning Herald	No	No	No	
The Market Online	No	Yes	No	
ABC News	Yes	Yes	No	Regular news mixed with financial news, difficult to use, untagged stock market
Small Caps	Yes	Yes	No	Sitemap highly conducive to scraping. Stock codes provided in articles. All about stocks — no general politics. Static web content (use BeautifulSoup)
Morning Star	No	Yes	No	No sitemap provided
Australian Stock Report	Yes	Yes	No	Unstructured in comparison to other article sites
Stock Head	Yes	Yes	No	Great sitemap but no list of stock codes in article
The Australian	No	No	No	
Hot Copper	No	Yes	No	

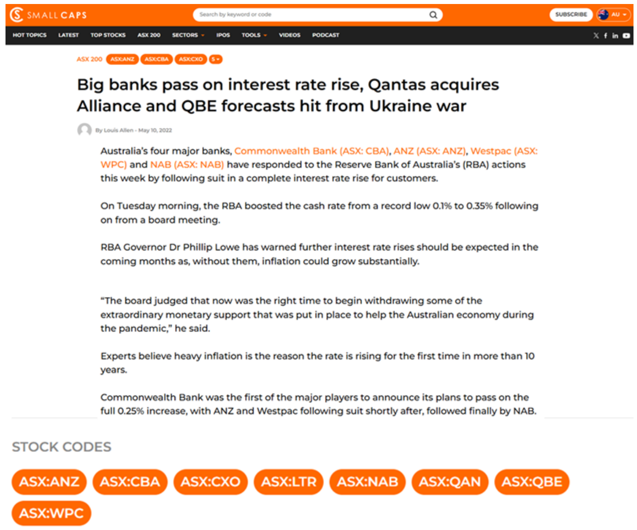
Table 1: Comparison of financial and market-related websites for article scraping suitability. Green rows indicate favourable conditions for scraping.

Article Year	**Number of Observations**
2023	10,468
2024	2,374
2025	323
Total	**13,165**

Table 2: Number of Observations per Year As reported by the XML Sitemap Last Modified Date

Now, individual article content needs to be scraped. The following listing uses `BeautifulSoup` and the `requests` library to extract the contents of a given URL, which was sourced from the sitemap. The `lxml` parser was chosen again for its speed, and since the necessary external C dependency had already been installed with the `lxml-xml` parser during the XML sitemap scraping, no additional setup was required [16]

The contents of the HTML were analysed and parsed into a dictionary capturing the title, author, date, sector, stock codes, and the article body, as demonstrated in the listing below. `BeautifulSoup` was used to extract data from the relevant tags. Below is an example output from the parsing process:



SMALL CAPS
Search by keyword or code
SUBSCRIBE AU
HOT TOPICS LATEST TOP STOCKS ASX 200 SECTORS IPOS TOOLS VIDEOS PODCAST

ASX 200 ASX:ANZ ASX:CBA ASX:CXO 5

Big banks pass on interest rate rise, Qantas acquires Alliance and QBE forecasts hit from Ukraine war

By Louis Allen - May 10, 2022

Australia's four major banks, Commonwealth Bank (ASX: CBA), ANZ (ASX: ANZ), Westpac (ASX: WPC) and NAB (ASX: NAB) have responded to the Reserve Bank of Australia's (RBA) actions this week by following suit in a complete interest rate rise for customers.

On Tuesday morning, the RBA boosted the cash rate from a record low 0.1% to 0.35% following on from a board meeting.

RBA Governor Dr Phillip Lowe has warned further interest rate rises should be expected in the coming months as, without them, inflation could grow substantially.

"The board judged that now was the right time to begin withdrawing some of the extraordinary monetary support that was put in place to help the Australian economy during the pandemic," he said.

Experts believe heavy inflation is the reason the rate is rising for the first time in more than 10 years.

Commonwealth Bank was the first of the major players to announce its plans to pass on the full 0.25% increase, with ANZ and Westpac following suit shortly after, followed finally by NAB.

STOCK CODES

ASX:ANZ ASX:CBA ASX:CXO ASX:LTR ASX:NAB ASX:QAN ASX:QBE ASX:WPC

Fig. 3. Example Article [2], containing fields such as title, author, date, and stock codes

Listing 1: Example HTML extraction code

```python
soup_test = extract_html_from_url('https://smallcaps.com.au/big-banks-interest-rate-rise-qantas-acquires-alliance-qbe-forecasts-hit/')
parsed_dict = parse_html_to_dict(soup_test)
parsed_dict
```

Key	Value
'title'	'Big banks pass on int...'
'author'	'Louis Allen'
'date'	'May 10, 2022'
'sector'	'ASX200'
'stock_codes'	['ASX: CBA', 'ASX: ANZ', ...]
'document'	"Australias four maj..."

Table 3: Parsed dictionary converted into a table

Exploratory Data Analysis

Table 3 depicts the structure of the data. Collected from smallcaps was 1532 articles. Figure 4 depicts the distribution of the scraped articles. The reports sector label is visualised in fig. 6, provided by the smallcaps website. The industry label is provided by the ASX website [3]. It is evident the high proportion of articles in the mining sector, potentially leading to bias in any investigation that is conducted in this data without any other intervention.

Analysing the textual corpus, fig. 5 shows the distribution of document length, where it can be seen that the mean document wordcount is 409 words, which combined with the quantity of data collected should be sufficient for training a machine learning model a source for this would be better than hopes and prayers

Feature Extraction

Sentiment Analysis

We model the sentiment of an article using the VADER (Valence Aware Dictionary and sEntiment Reasoner), as this has shown to having promising results in a similar setting, as found by [17]. This implementation of vader is assisted through a Geeks for Geeks instructional article [18]

VADER's rule-based approach relies on fixed lexicons and doesn't fully capture the context-dependent

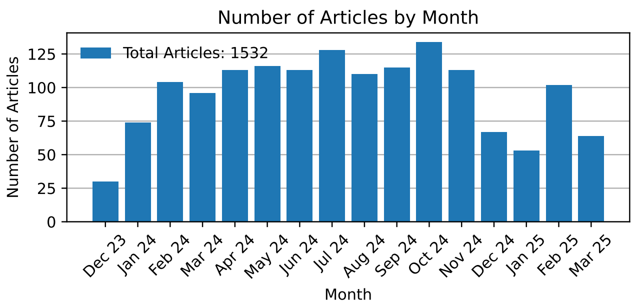


Fig. 4. Quantity of Scraped Articles per Month

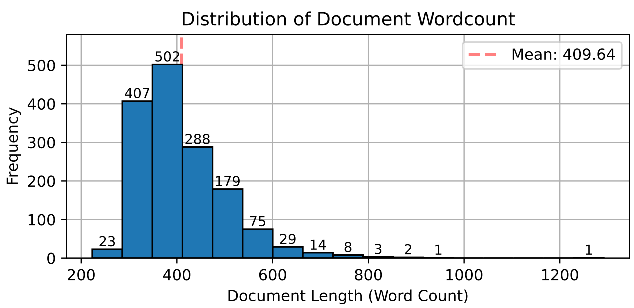


Fig. 5. Distribution of Document Length

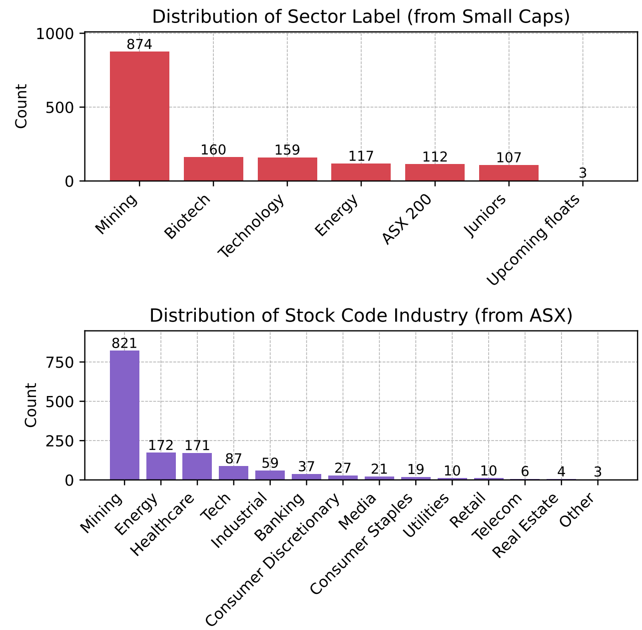


Fig. 6. Distribution of Articles by Sector, using the *smallcaps* industry label and the ASX sector label [3]

sentiment embedded in text, whereas BERT's deep contextual embeddings allow it to discern and represent nuanced sentiment more effectively. FinBERT is a domain-specific adaptation of the BERT model, fine-tuned on financial texts to accurately determine the sentiment expressed in this particular context [19]. This model was retrieved using the huggingface library [20]

While this sounds better, it relies on the data being labelled, making training on a large corupus of data unfeasible. We therefore utilise the finbert-tone pretrained transformer which is the FinBERT model trained on 10,000 manually annotated documents (positive, negative, neutral) sentences from analyst reports.

Machine Learning

After having sraped the articles for the documents and sentiments? a machine learning model was developed to predict stock price. figs. 7 and 8 show the structure of a typical feed forward neural network and a recurrent neural network. Recurrent neural networks (RNNS), and others like RNNS, are particularly well suited to sie seues data

[17] show that an LSTM model, when trained on timeseries price data and sentiment data from news articles, can outperform a model trained on price data alone. Analysis of the literature suggests that SVM models, whilst the most popular [1] often underperformed in a financial setting, due to parameter optimisation being required [1, 21]

Talk about the adam optimiser According to Kingma et al., 2014, the method is "computationally efficient, has little memory requirement, invariant to diagonal rescaling of gradients, and is well suited for problems that are large in terms of data/parameters". [22, 23]

the events extracted from the news are usually quite sparse, resulting in unreliable and unsustainable predictive power [21] (bar for bar) therefore we pivot and try and predict on the asx index which should address saprsity issues, rather than trying to predict voltaile single stocks

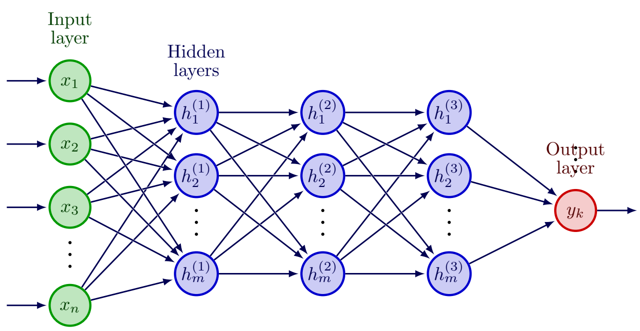


Fig. 7. Typical "FeedForward" Neural Network Topology [4]

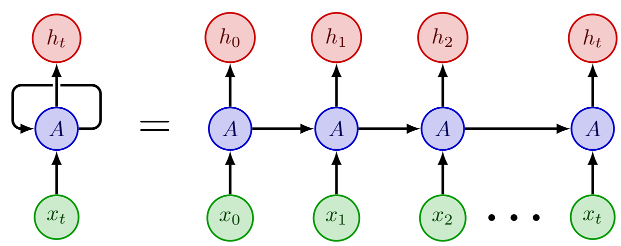


Fig. 8. Recurrent Neural Network Topology [5]

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